Sahaj Bhadja

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Summary

Software engineer with 2+ years building customer-facing mobile and web products, REST APIs, and reliable cloud workflows. Experienced with React Native, TypeScript, Python, Firebase, testing, secure sessions, and production delivery across iOS, Android, and web.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected Azure SQL and Blob Storage services for 500+ users, tuning indexed queries and secure access patterns to reduce customer-facing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI workflows for 1,000+ monthly jobs using explicit state transitions, retries, and failure isolation, tripling throughput while preserving reliable production behavior under concurrent load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Delivered React interfaces and Flask REST APIs for secure file upload, grading history, and asynchronous feedback workflows, supporting 150+ instructors and 10,000+ daily production requests.
- Designed concurrent processing for 1,000+ uploaded files with responsive job-state updates and fault-tolerant execution, reducing feedback turnaround 60% while maintaining stable API performance during peak demand.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable PyTorch simulation pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through reproducible experiments, structured evaluation, and systematic debugging of model behavior.
- Developed a spatial-temporal encoder integrating map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating behavior across simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform microservices with 20+ automated unit and integration suites using PyTest, Mockito, and Jenkins, validating releases across Windows, Linux, and macOS production environments.
- Optimized REST integrations using object-oriented patterns, AWS Glue, S3, filtering, and pagination, improving response times 30% for bandwidth-constrained clients and increasing ETL pipeline efficiency 68%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built React Native clients for iOS and Android with Firebase persistence, real-time booking updates, image storage, notifications, and session checks across a service platform handling 50,000+ records.
- Developed Flask REST APIs for bookings, pricing, coupons, and payments, delivering sub-200-millisecond responses and webhook-driven synchronization that improved booking confirmation speed 40% across mobile experiences.

Skills

Mobile & Frontend: React Native, iOS, Android, React, TypeScript, JavaScript, HTML5, CSS3, responsive UI, mobile notifications
Backend & APIs: Python, Flask, FastAPI, Node.js, REST APIs, JSON, WebSockets, microservices, asynchronous workflows
Data & Cloud: Firebase, Firestore, PostgreSQL, MySQL, SQL Server, Azure SQL, AWS, Azure, S3, Blob Storage
Quality & Delivery: Git, GitHub Actions, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, debugging, Agile, Scrum, secure sessions, production support

Projects

HMIS | *TypeScript, MERN, REST APIs, Appwrite, Twilio, Sentry* | Live
- Built responsive appointment and patient-record workflows with consent-based API access, secure authentication, SMS notifications, and Sentry monitoring, improving usability and reliability across user-facing healthcare experiences.

FinNexus | *Next.js, TypeScript, WebSockets, MongoDB, Plaid, Kubernetes* | Live
- Delivered a responsive financial interface supporting 50,000+ users and one million transactions, integrating secure bank linking, real-time state synchronization, advanced filtering, and containerized production deployment.

PiXMod | *Next.js, TypeScript, MongoDB, Stripe, CLIP, SAM* | Live
- Created a responsive AI image-editing product with segmentation, semantic search, subscriptions, and scalable metadata storage, processing 1,000+ edited assets through polished customer-facing workflows.